Market Inefficiencies Surrounding Energy Announcements

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Abstract

We use sequential energy inventory announcements to shed new light on the informational efficiency of financial markets. Our findings provide clear evidence of inefficiency in crude oil futures and stock markets. This inefficiency can be exploited by sophisticated traders. We examine the effect of market liquidity on the efficient incorporation of information in this setting. We also construct a predictor that can predict inventory surprises and pre-announcement returns in-sample and out-of-sample. Finally, we develop a combination forecast that can be used as a proxy for market expectations of oil inventory announcements.

JEL classification: C53; D81; G4; G14; G17; Q474

Keywords: Inventory Announcements, Market Efficiency, Financial Market Predictability, Oil Futures

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1. Introduction

Oil inventory announcements move energy markets. These announcements provide essential signals regarding the supply and demand of oil. Two of the primary energy announcements, made by the American Petroleum Institute (API) and the U.S. Energy Information Administration (EIA), occur weekly. These two public announcements provide very similar fundamental information, and they are closely followed by traders. According to the efficient market hypothesis, fundamental public information should be immediately impounded into asset prices. The immediate price adjustment prevents traders from exploiting public information to make profitable trades (French and Roll, 1986).

There is no consensus regarding the efficiency of energy futures markets. Shambora and Rossiter (2007) use an artificial neural network to analyze crude oil futures prices. They construct a profitable technical trading strategy, which suggests that the oil futures market is not informationally efficient. Wang and Yang (2010) use intraday data for crude oil, heating oil, gasoline, and natural gas futures and test the efficiency of these markets using several nonlinear models. They find that the crude oil and gasoline futures markets are efficient, but heating oil and natural gas futures show weak-form efficiency only during the bull market.[1] Kristoufek and Vosvrda (2014) investigate the market efficiency in 25 commodity futures. They find WTI crude oil, heating oil, and coffee to be the most efficient compared to the other commodities. Our study contributes to this debate by examining whether crude oil futures and oil industry stocks immediately incorporate information contained in oil inventory announcements.

The API oil inventory announcements are followed 18 hours later by the oil inventory announcements from the EIA. We use the unique sequential nature of these events to analyze the

[1] In a weak-form efficient market, past market data cannot be used to forecast future returns.

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efficiency of oil futures markets and to predict oil futures returns before the EIA oil announcements. We find that the oil futures return after the API announcements significantly predicts the return before the EIA announcements. This predictability of the pre-EIA oil futures return provides strong evidence that oil futures do not immediately incorporate all of the information released by the API.

Chordia, Roll, and Subrahmanyam (2008), examine the role of liquidity in market efficiency. They find that the predictability of returns from past order flows decreases when liquidity is high. Furthermore, higher trading activity of algorithmic and high-frequency traders facilitates liquidity and price discovery (Brogaard et al., 2014; Hasbrouck & Saar, 2013). Therefore, high liquidity should facilitate the efficient incorporation of information in oil futures prices. Consistent with this expectation, we find that the oil futures returns after the API oil announcements significantly predict the pre-EIA announcement returns only during periods of relatively low liquidity.

Furthermore, we contribute to the debate regarding whether changes in oil prices can predict stock returns. Huang et al. (1996) find that oil futures are not correlated with or linked to the stock market, except stocks of oil companies. However, others have found that oil has a significant impact on stock markets depending on the type of oil shock (Chiang, Hughen, and Sagi, 2015; Kilian and Park, 2009; Ready, 2018), and the specific sectors of the stock markets (Fan and Jahan-Parvar, 2012). We find that the opening stock prices of oil companies do not fully incorporate oil inventory information released after trading hours. The post-API announcement oil futures return is a significant predictor of the returns of oil companies on the following day.

The predictive ability of the post-API oil futures return for the pre-EIA oil futures return suggests the importance of the information released by the API in shaping the market expectations

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for the EIA oil announcements the following day. Most of the literature evaluates the impact of EIA announcements by considering a single proxy for market expectation; for example, Bloomberg median consensus, (Wolfe and Rosenman, 2014), Reuters forecast (Bu, 2014), or the API actual (Armstrong, Cardella, and Sabah, 2021). We construct a predictor based on the difference between the API actual and the Bloomberg consensus forecast. The predictor has significant explanatory power over the pre-EIA announcement return. In addition, a simple trading strategy based on the predictor would generate an annualized Sharpe ratio of 0.67 in the full sample and an annualized Sharpe ratio of 2.1 in a subset of the sample with values of the predictor in the top and bottom deciles.

Finally, we employ the relative importance method to construct a combination forecast of the EIA oil announcements by objectively allocating weights to the Bloomberg median consensus and API actual (Grömping, 2006; Thomas, Zhu, Zumbo, and Dutta, 2017). Using our combination forecast as a proxy for market expectations increases the R^2 in the regression of oil futures returns after the EIA announcements on inventory surprises by approximately 130% compared to the Bloomberg consensus forecast, and by approximately 17% compared to the API actual. Hence, using the combination forecast to compute oil inventory surprises reduces measurement errors that could stem from the use of a single measure of market expectations.

Two pevious studies show evidence of informed trading before energy inventory announcements. Rousse and Sévi (2019) find some "suspicious trading" patterns (i.e., significant order imbalances in the "right" direction) before the EIA oil inventory announcements. They attribute this pattern to information leakage. Our paper differs from theirs in several important aspects. First, our findings point towards sophisticated trading that is driven by mere public information which includes the API announcement released the day before. Second, we investigate

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sophisticated trading around both EIA and API announcements. Our findings indicate that the significant pre-announcement drift exists only before the EIA release. This provides evidence that some traders have superior skills for processing public information.

In a related paper, Gu and Kurov (2018) find evidence of informed trading, before the Weekly Natural Gas Storage Report. The difference between forecasts of analysts with superior historical forecasting ability and the Bloomberg consensus forecast predicts inventory surprises and the pre-announcement returns of natural gas futures. Our paper differs from Gu and Kurov's (2018) study in several ways. First, we show that the oil futures return following the API inventory announcements is a significant predictor of the pre-EIA announcement returns. This is a clear violation of weak-form market efficiency. Second, we examine whether liquidity influences the return predictability around the release of the Weekly Petroleum Status Report. Third, in addition to looking at the efficiency of the oil futures market we show evidence of inefficiency in stocks of oil companies. Specifically, opening prices of energy stocks do not fully incorporate public information released after the market close the day before.

We make several contributions to the literature. First, we provide clear evidence of weak-form market inefficiency in the oil futures market and in the oil industry stocks. Second, we show that mere public information can be used to predict oil futures returns before the EIA announcements in-sample and out-of-sample, and to predict returns of oil industry stocks. This alleviates the concerns about possible preannouncement information leakage (Rousse & Sévi, 2019). Finally, we show that a simple trading strategy based on the API announcements and the Bloomberg consensus forecasts delivers a high annualized Sharpe ratio.

The remainder of the paper is organized as follows. The next section describes the institutional background, data and key variables. Section 3 discusses the empirical results. Section

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4 briefly concludes the paper.

2. Institutional Background, Data, and Variables

Our empirical analysis centers around two major weekly oil announcements. These two announcements provide a unique setting because they contain similar information and are released on consecutive days. One of these announcements is released by the EIA. Oil companies and refineries are required to participate in the weekly EIA survey and provide accurate information about their oil inventory as of 7:00 a.m. each Friday.[2] The EIA compiles this information and prepares the Weekly Petroleum Status Report. The report is usually released on Wednesdays at 10:30 am ET. Market participants and policymakers closely follow these releases and use them to analyze changes in the oil supply and demand.

The American Petroleum Institute (API) was established by Congress and the domestic oil and gas industry in 1919. The institute has around 625 corporate members, and it is considered one of the largest associations in the oil and gas industry.[3] The API releases its oil inventory report at 4:30 pm ET every Tuesday. These releases are reported by the traditional news media, social media and other media platforms (Armstrong et al., 2021). Prior research finds that the API oil inventory announcements have significant effects on oil prices and volatility (Ye and Karali, 2016). The API and the EIA provide similar information, and both releases are considered highly reliable. Furthermore, both EIA and API collect data from almost the same respondents and cover around 90% of the U.S. oil and gas industry. According to API, "API collects an exact copy of the data submitted to EIA. Respondents send data to API using the same weekly survey forms that EIA uses."

[2] The EIA information is provided on the EIA's website at https://www.eia.gov/.
[3] The API information is available on the API's website at https://www.api.org/.

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Our sample period spans from January 2011 to August 2019.[4] This sample period provides sufficient data to conduct our analysis and ensure that our results are not driven by a particular market regime. Due to holidays, the EIA announcements are occasionally released at 11:00 a.m. on Thursdays instead of 10:30 a.m. on Wednesdays. In such cases, the EIA announcements are released 30 minutes after the Natural Gas Storage Report, which is released at 10:30 a.m. on Thursdays. Since the natural gas inventory announcements may have a confounding effect on our results, in our main analysis, we exclude those EIA announcements that took place on Thursdays.[5] The resulting sample contains 385 EIA oil inventory announcements.

We use data for the nearby WTI oil futures contracts, since they are very liquid (Chiang and Hughen, 2017). These futures are considered to be very efficient in incorporating public information. We use intraday data, which allows us to analyze trading activity and price behavior around the EIA and API inventory announcements. Figure 1 provides a visual illustration of the average level of trading activity in the nearby WTI crude oil futures in the 24-hour window containing the API and EIA oil inventory announcements. Trading activity declines substantially after the daily settlement price is determined 2:30 p.m. After a brief spike in trading after the API announcement at 4:30 p.m., trading volume declines again and trading is halted for an hour from 5:00 p.m. to 6:00 p.m. Trading activity remains low until the following morning, increases substantially at 9:00 a.m., and then briefly jumps by a factor of almost seven to about 4,000 contracts per minute right after the EIA inventory announcement.

[Insert Figure 1 here]

We use the oil futures data to construct the illiquidity measure following Amihud (2002). This illiquidity measure is computed as follows:

[4] The sample period is chosen based on availability of the API actual data.
[5] Our results remain significant and very similar if these announcements are included in the sample.

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$$ILLIQ = \left(\frac{|r_{d,s}|}{vol_{d,s}}\right) \quad (1)$$

where $r_{d,s}$ is the absolute return and $vol_{d,s}$ is the trading volume on day d in period s.[6] This measure is widely used in the literature (Goyenko, Holden, and Trzcinka, 2009).[7]

To measure inventory news, it is important to choose an appropriate proxy for market expectations. We employ the relative importance method to construct a combination forecast which could accurately represent the market expectations. This methodology has been employed in different contexts to assess the relative importance of multiple variables (Grömping, 2006; Johnson and Lebreton, 2004; Thomas et al., 2017). To the best of our knowledge, the only prior study that uses combination forecasts to compute oil inventory surprises is Anatolyev, Seleznev, and Selezneva (2018). Our approach differs from theirs in that we do not impose any assumptions in allocating forecast weights. Furthermore, our approach is easily generalized to incorporate more than two forecasts. This approach to measuring market expectations can be used in other contexts such as macroeconomic and corporate announcements.

We construct the combination forecast according to the weights estimated by the relative importance analysis. This method is proposed by Lindeman, Merenda, and Gold (1980), henceforth LMG. Figure 2 shows the annual weights allocated to the API actuals computed based on relative importance regression using data for the previous year. We can see that the API actual receives weights of more than 50% throughout our sample period. Table 1 provides summary statistics for the main variables used in the analysis. The table shows that the means of both API actual and Bloomberg consensus are below the mean of the EIA actual. In addition, the mean of

[6] The illiquidity is measured between 4:30 p.m. ET on Tuesday and 10:30 a.m. on the following day.
[7] The Amihud measure has been shown to have the largest correlation with liquidity benchmarks in commodity markets (Marshall, Nguyen, and Visaltanachoti, 2012).

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the surprises that are computed based on the combination forecast falls between the means of the surprises computed based on Bloomberg consensus and API actual.

[Insert Figure 2 and Table 1 here]

3. Results and Discussion

As mentioned above, we use the sequence of two very similar U.S. oil inventory announcements to examine the informational efficiency of energy markets. Specifically, we examine whether some information contained in the API announcements is incorporated in prices of oil futures and energy stocks during the following morning, shortly before the EIA announcements. In this section, we discuss our results and relate them to the existing literature. Rousse and Sévi (2019) find a significant drift in the "correct" direction in the crude oil futures market before the EIA oil inventory announcements. They attribute this drift potentially to information leakage. Gu and Kurov (2018) ascribe a similar drift before the EIA natural gas inventory announcements to superior forecasting. Consistent with these studies, Panel A of Figure 3 shows a price drift in the oil futures market before the EIA oil inventory announcements. Below, we provide evidence that processing mere public information explains much of this preannouncement drift. Panel B of Figure 3 shows no evidence of statistically significant drift prior to the API oil inventory announcements, which is consistent with the API oil inventory surprises being less predictable compared to the EIA inventory surprises based on the Bloomberg consensus forecast.[8]

[Insert Figure 3 here]

[8] We use the Thomson Reuters survey as a proxy for market expectations for the API announcements because the results of this survey are usually released at about 2 pm ET on the announcement days (Ye and Karali, 2016). The cumulative average returns are very similar when we use the Bloomberg consensus as a proxy for market expectations.

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3.1. Predicting Returns Before the EIA Inventory Announcements

We begin by testing whether the information contained in the API inventory announcements is efficiently incorporated in oil futures prices. The efficient market hypothesis posits that prices reflect all public information. Therefore, earlier returns should not predict subsequent returns. Panel A of Table 2 shows that the 30-minute return of oil futures after the API release predicts the 60-minute return of oil futures prior to the EIA oil inventory announcements.[9] We find almost the same significant predictability of the oil futures returns in other time intervals before the EIA oil inventory announcements such as 30 and 90 minutes before the announcement.[10] We also find no evidence of significant asymmetry in the predictability of the pre-EIA oil futures return. Specifically, the difference between the coefficient estimates of positive post-API oil futures returns and negative post-API oil futures returns in the predictive regression for the pre-EIA oil futures return is not statistically significant. To account for a potential effect of outliers, we re-estimate the regression using the outlier-robust procedure of Yohai (1987). The robust regression results reported in Table 2 are very similar to the OLS estimates.

[Insert Table 2 here]

Figure 4 uses Welch and Goyal (2008) methodology to show significant out-of-sample predictive power of the post-API oil futures returns for the pre-EIA returns.[11] These findings present an apparent violation of weak-form market efficiency. The information in the API announcements is not fully incorporated in the price immediately, perhaps because some traders delay trading on this information until right before the EIA announcement. Thus, our results

[9] We use 30-minute return of oil futures after the API announcement because these announcements are released at 4:30 pm ET and WTI crude oil futures do not trade from 5:00 p.m. to 6:00 p.m. ET.
[10] We use the 60-minutes interval before the EIA announcements to conduct our main analysis because it represents a midpoint between these alternatives.
[11] An explanation of the Welch and Goyal (2008) methodology is provided in the Appendix.

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indicate that the price drift before the EIA oil announcements documented by Rousse and Sévi (2019) can be explained by delayed adjustment to public information.

[Insert Figure 4 here]

The delayed incorporation of information released by the API may be related to the sequential nature of the two inventory announcements and the relatively low futures market liquidity after the API announcement. The API oil inventory announcements are known to provide reliable estimates of the U.S. oil inventory level. However, the EIA announcements are more widely recognized and followed compared to the API announcements. Furthermore, WTI crude oil futures have a one-hour trading break starting at 5:00 p.m. ET. As shown in Figure 1, when trading resumes at 6:00 p.m. the market liquidity is low. These factors may contribute to delayed incorporation of information in the API oil inventory announcements.

The Bloomberg consensus forecast of oil inventory changes is publicly available before the inventory announcements. However, this forecast is almost never updated after the API announcements. Since this forecast does not reflect the API oil inventory news, it is not an up-to-date measure of market expectations at the time of the EIA oil inventory announcements. Hence, we construct a simple predictor computed as the difference between the API actual and the Bloomberg median consensus. The predictor represents a forecast of the surprise of the EIA oil inventory announcements, if one uses the widely followed Bloomberg consensus as a proxy for market expectations. Panel B of Table 2 and Figure 5 show that the predictor can predict the oil futures returns before the EIA announcements both in-sample and out-of-sample. These results provide further evidence of market inefficiency because the predictor is based on public information.[12]

[12] As part of our robustness checks, we split the sample in two equal sub-samples and repeated the entire analysis in Table 2. The results in both subsamples are similar to those reported in the paper.

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[Insert Figure 5 here]

3.2. Predictability of Inventory Surprises

We further analyze whether the EIA inventory surprises can be predicted using the post-API oil futures return and our predictor. Following previous studies (e.g., Halova, Kurov, and Kucher, 2014), we compute the oil inventory surprises as the difference between the EIA actual and the Bloomberg median consensus divided by the level of inventory. Table 3 shows that both the oil futures returns after the API announcements and the predictor can predict the oil inventory surprises. Hence, the EIA oil inventory surprises are highly predictable by using public information, probably due to the fact that the Bloomberg forecasts of the EIA oil inventory changes are rarely updated after the API inventory releases. To address the potential autocorrelation in the EIA inventory surprises, we include lags of the surprises in the model.[13] Table 3 shows that this leaves our results essentially unchanged.

[Insert Table 3 here]

3.3. Market Liquidity and Return Predictability

Elaborating on our primary findings under different market conditions, Table 4 provides the results of our analysis during times with different levels of liquidity. Liquidity plays a vital role in market efficiency. Green (2004) shows that price discovery surrounding the release of public announcements occurs through trading. Therefore, information will be incorporated into prices faster when the market is more liquid. Several recent studies show that sophisticated trading, such as high-frequency and algorithmic trading, enhances price discovery and liquidity (Brogaard et al., 2014; Hasbrouck and Saar, 2013). Accordingly, we expect that as liquidity increases, return

[13] The optimal numbers of lags were determined using the Akaike information criterion.

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predictability driven by market inefficiencies declines. Consistent with this view, our results in Table 4 show that the predictive power of the post-API oil futures return for the pre-EIA oil return is significant only when the liquidity in the oil futures market is relatively low.

[Insert Table 4 here]

3.4. Return Predictability for Energy Stocks

We also examine if a similar market inefficiency is present in energy stocks. Changes in oil prices represent important fundamental information for energy companies. The regression estimates in Table 5 show that the 30-minute post-API crude oil futures return can be used to predict the returns of oil companies from the CRSP database on the day of the EIA oil inventory announcements.[14] This suggests that opening prices of energy stocks do not fully incorporate information about oil inventory released after the previous stock market close.

We also use intraday data for the NYSE ARCA Oil and Gas Index to examine how long it takes for energy stocks to incorporate information contained in the API oil inventory announcements. Table 5 shows that energy stocks incorporate this information within the first ten minutes of the stock market opening. This predictability is no longer significant after the first ten minutes of trading. Our findings support the view that oil has predictive ability for stock returns. In addition, the 30-minute post-EIA oil futures return has no significant predictive power over the NYSE ARCA Oil and Gas Index return during the time interval from 11:00 am until the stock market close.[15] Therefore, energy stocks seem to be efficiently incorporating oil-related fundamental information during the regular trading hours.

[Insert Table 5 here]

[14] We identify oil companies using the following SIC codes: 1300, 1310-1319, 1320-1329, 1330-1339, 1370-1379, 1380-1382, 1389, 2900-2912, 2990-2999, 3533, 4612-4613, 5171-5172, 6792.
[15] These results are not tabulated for brevity but are available upon request.

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3.5. Combination Forecast

As mentioned above, the Bloomberg consensus forecast does not capture changes in market expectations after the API inventory announcements. However, this does not rule out the possibility that the Bloomberg consensus forecast contains information not captured by the API actual. Regression estimates in Table 6 show that both the Bloomberg median forecast and the API actual are useful predictors of the weekly EIA inventory change.

[Insert Table 6 here]

We want to explore how the choice of a proxy for market expectations influences the estimated impact of the EIA oil inventory surprises on the oil futures returns. Panel A of Table 7 shows that using the Bloomberg consensus forecast as the market expectations proxy yields an estimated OLS response coefficient of -0.24 and an R^2 of 9%. In contrast, computing the EIA oil inventory surprises as the difference between the EIA and API actual announced values produces an estimated market response coefficient of -0.42 and an R^2 of 18%.[16] The robust regression estimates in Panel B of Table 7 are similar to the OLS results. These results validate the importance of using the information released by the API in measuring market expectations prior to the EIA announcements. Based on this analysis, we employ a relative importance method to construct a combination forecast of the EIA oil inventory announcements by objectively allocating weights to the Bloomberg consensus forecast and the API actual.

[Insert Table 7 here]

Several studies in the finance literature have used combination forecasts (Baumeister & Kilian, 2015; Detzel & Strauss, 2018; Rapach et al., 2010). These studies provide evidence that combination forecasts deliver substantial gains in forecasting accuracy. To the best of our

[16] To compute the surprises, we scale the unexpected inventory changes by the level of inventory as in Halova, Kurov and Kucher (2014).

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knowledge, the only prior study that uses combination forecasts to compute oil inventory surprises is Anatolyev, Seleznev, and Selezneva (2018). By using the relative importance method, we do not need to impose any assumptions in allocating forecast weights. Furthermore, the relative importance method can be generalized to incorporate more than two forecasts. This approach to measuring market expectations can be used in other contexts such as macroeconomic and corporate announcements. Grömping (2006) provides a discussion of this approach.

The LMG method decomposes the R^2 of a multivariate regression into non-negative contributions of each independent variable. Furthermore, the LMG technique overcomes the potential problem that the contribution of each regressor depends on the order in which the regressors are added to the model. The LMG method achieves this by using simple averaging over all the possible orderings of the regressors. To describe the LMG approach, we use the same notations and steps as in Grömping (2006). For a set *S* that consist of all regressors in the model, the R^2 can be expressed as:

$$R^2(S) = \frac{Model\ SS\ (model\ with\ regressors\ in\ S)}{Total\ SS} \tag{2}$$

Once we add an additional regressor that is in set *M* to the previous set *S*, the additional R^2 can be computed as follows:

$$seqR^2\ (M|S) = R^2\ (M \cup S) - R^2(S) \tag{3}$$

The number of possible orderings is determined by the permutations of the chosen regressors $x_1, \dots, x_p$ which are denoted by the tuple of indices $r = (r_1, \dots, r_p)$. The model is assumed to include multiple regressors, $S_k(r)$, before the regressor x_k. The part of the R^2 allocated to the regressor x_k can be formulated as:

$$seqR^2\ (\{x_k\}|S_k(r)) = R^2\ (\{x_k\} \cup S_k(r)) - R^2(S_k(r)) \tag{4}$$

Finally, from equation (3) we can get the LMG metric as follows:

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$$LMG\ (x_k) = \frac{1}{p!} \sum_{r\ permutation} seqR^2(\{x_k\}|r) \quad (5)$$

We compute annual weights of the API actual and the Bloomberg median consensus in the combination forecast using data for the previous year. Panel A of Table 7 shows that using the combination forecast as a proxy for market expectations increases the R^2 of the energy announcement impact on oil futures return by approximately 130% compared to the Bloomberg median consensus, and by approximately 17% compared to the API actual. Hence, using the combination forecast to compute the oil inventory surprises reduces measurement errors that could stem from the use of a single proxy for market expectations.[17]

3.6. Identification-Through-Censoring

To analyze the effect of measurement error of the estimated market response coefficients, we conduct an analysis using the identification-through-censoring (ITC) technique proposed by Rigobon and Sack (2008).[18] Regressors are often measured with error. It is well known that this measurement error induces a downward bias in the regression coefficients estimated with common techniques. The ITC methodology adjusts for such attenuation bias. Oil inventory surprises contain measurement errors for a number of reasons. Any proxy for expectations does not incorporate all available information. More specifically, the EIA surveys do not cover the entire population of oil companies. Halova, Kurov, and Kucher (2014) employ the ITC technique to correct for biases in the estimates of the responses of energy futures markets to inventory news. They find that the attenuation bias in the OLS estimates is quite large.

The ITC estimates of the market response coefficients are reported in Panel C of Table 7.

[17] We have explored other proxies for market expectations before the EIA announcements, including the Bloomberg top-ranked analyst forecast. We used the Bloomberg consensus in weeks with missing top-ranked analyst forecast. The results are almost the same as those obtained using the Bloomberg consensus as a proxy for market expectations.
[18] A more detailed description of the ITC technique is provided in the Appendix.

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For all three measures of market expectations, the ITC estimates are close to -2. These estimates are much larger in absolute value than the corresponding OLS and robust regression estimates reported in the same table. These estimates are also about twice as large as the ITC estimate of the oil futures market response to crude oil inventory news in an earlier sample period reported in Halova, Kurov, and Kucher (2014). The larger ITC coefficient estimates in our study could reflect the rising importance of the U.S. in the energy markets. Due to the increased production of shale oil, the U.S. has become one of the largest oil producers in the world.

Table 7 also shows that when we use the combination forecast as a measure of market expectations, the OLS estimate of the market response coefficient is closer to the estimates produced by the ITC technique compared to the OLS estimates based on either the Bloomberg median consensus or the API actual. Although the estimated proportion of the variance of the measured inventory surprise due to noise is quite large for all three measures of market expectations, it is smaller for the surprise based on the combination forecast compared to the surprises based on the other two expectation proxies. These results provide evidence that using the combination forecast to compute oil inventory surprises reduces measurement errors that stem from the use of a single stale proxy for market expectations.

3.7. Economic Significance

To examine if oil futures return predictability before the EIA oil inventory announcements can be used for profitable trading, we propose a simple trading strategy based on our predictor, computed as the difference between the API actual announced value of inventory change and the Bloomberg consensus forecast. The strategy involves selling (buying) oil futures contracts 60 minutes before the EIA announcements and closing the position one minute before the EIA announcement if the predictor is positive (negative). Table 8 shows that this trading strategy generates an annualized

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Sharpe ratio of about 0.67 in the full sample, and an annualized Sharpe ratio of 2.1 when the predictor has large values. The large values of the predictor are those in the top and bottom deciles.[19]

[Insert Table 8 here]

Since events with large values of the predictor occur about nine times a year on average, the annualized Sharpe ratio is high, but not extraordinarily high for a strategy that involves frequent trading. For example, Wen, Gong, Ma, and Xu (2021) report a Sharpe ratio of about nine for an intraday market timing strategy based on predictability of returns of the United States Oil Fund ETF. Menkveld (2013) estimates an annualized Sharpe ratio of over 20 for a high-frequency trading firm. We assume traders would open a futures position 60 minutes before the EIA announcements to be consistent with the other analyses in our study. Table 8 also shows that this trading strategy generates substantial Sharpe ratios if it involves establishing a position 90 or 120 minutes before the EIA announcements.

We also examined several trading strategies that include opening a futures position shortly before the EIA announcement and closing it after the announcement. The Sharpe ratios for these strategies are lower than those in Table 8. A potential reason for the lower Sharpe ratios is that information used to compute the predictor (the API actual and the Bloomberg consensus forecast) is publicly available before the EIA announcement. Multiple traders probably trade on this predictable component of the EIA announcement. As a result, this information is fully impounded in futures prices before the EIA announcement. Gu and Kurov (2018) report a similar finding for

[19] To make sure the results are not driven by a few outliers, we repeated this analysis after removing observations with the absolute value of the predictor in the top 1% (in the whole sample of 385 announcements) from the top and bottom deciles. The resulting 72 observations produced an annualized Sharpe ratio of 2.01.

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the natural gas futures around the Natural Gas Storage Report announcements, although their predictor is constructed using different information.

4. Conclusion

This paper uses sequential energy announcements to examine how futures and equity markets process information. We find that the oil futures return after the weekly inventory releases by the American Petroleum Institute (API) predicts the oil futures return before the U.S. Energy Information Administration (EIA) announcements on the following day. This predictability is statistically significant during periods of relatively low liquidity. Additionally, we find that the API actual and Bloomberg consensus forecasts are both useful in forecasting the change in oil inventory released by the EIA. We construct a predictor computed as the difference between the API actual and the Bloomberg median consensus. The predictor has significant explanatory power for oil futures returns before the EIA announcements both in-sample and out-of-sample. A simple trading strategy using large values of the predictor to generate trading signals would produce an annualized Sharpe ratio of 2.1.

We develop a combination forecast based on the API actual and the Bloomberg median consensus to forecast the EIA oil announcements. The combination forecast increases the R^2 of the regression of oil futures returns on energy surprises by approximately 130% compared to the Bloomberg median consensus used in prior studies. Overall, our results show that asset prices do not instantaneously reflect new public information even in the case of widely anticipated scheduled announcements. Our results are useful for traders. Policymakers tasked with facilitating efficient price discovery in energy futures markets may also benefit from our study. For example, our results suggest that public announcements are more likely to be incorporated immediately into asset prices if they are released when the market liquidity is high.

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Data Availability Statement

The data used in this study are available from Genesis Financial Technologies, CRSP database, Bloomberg and the Energy Information Administration (EIA). Restrictions apply to the availability of futures market data, CRSP stock return data and Bloomberg EIA announcement data. These data sets can be acquired by paying subscription fees. The U.S. crude oil inventory level data is freely available on the EIA website. The API crude oil inventory changes are freely available at https://www.investing.com/.

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Data Citation

[dataset] Intraday data for nearby WTI crude oil futures and the NYSE ARCA Oil and Gas Index; 2011-2019; not publicly available; data can be obtained from Genesis Financial Technologies on payment of a data service fee; https://www.tradenavigator.com/.

[dataset] Daily data for U.S. oil companies; 2011-2019; not publicly available; data can be obtained from the Center for Research in Security Prices (CRSP) database on payment of a subscription fee; http://www.crsp.org/.

[dataset] Actual values and median forecasts of the DOE U.S. crude oil inventory changes; 2011-2019; not publicly available; data can be obtained from a Bloomberg terminal on payment of a subscription fee; https://bba.bloomberg.net/.

[dataset] U.S. crude oil inventory levels; 2011-2019; U.S. Energy Information Administration; https://www.eia.gov/petroleum/data.php.

[dataset] U.S. crude oil inventory changes reported by the American Petroleum Institute; 2011-2019; https://www.investing.com/.

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Table 1: Summary Statistics

Variable	Mean	St. Deviation	Minimum	Maximum
EIA Actual	307.81	4588.46	-12788	14400
API Actual	191.11	4614.72	-12400	14200
Predictor	-0.01	1.02	-3.34	2.86
Combination Forecast	199.38	3703.36	-9560.56	13116.67
Post-API Return	0.003	0.51	-2.36	2.01
Bloomberg Median Consensus	178.74	2256.5	-4261	2240
$\text{Surprise}^{\text{Bloomberg}}$	0.025	0.98	-2.80	2.75
$\text{Surprise}^{\text{API}}$	0.031	0.76	-1.99	2.06
$\text{Surprise}^{\text{Combination}}$	0.026	0.70	-2.31	1.88

This table reports summary statistics for the main variables used in the analysis. The EIA actual, API actual, combination forecast, and Bloomberg median consensus are in thousands of barrels. The predictor is computed as the difference between the API Actual and the Bloomberg median consensus divided by the level of inventory and is expressed in percentage terms. The combination forecast is computed as the weighted average of the Bloomberg median consensus and the API actual, with the weights computed using the relative importance methodology described in Section 3. Post-API Return is the oil futures return in the 30 minutes after the API announcement. $\text{Surprise}^{\text{Bloomberg}}$ is the oil inventory surprise computed using the Bloomberg consensus, $\text{Surprise}^{\text{API}}$ is the oil inventory surprise computed using the API actual. $\text{Surprise}^{\text{Combination}}$ is the oil inventory surprise computed using the combination forecast. The inventory surprises are expressed in percentage and computed as the difference between the EIA actual and the corresponding market expectations proxy divided by the inventory level. The sample period is from January 2011 to August 2019 and contains 385 announcements.

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Table 2: In-sample Predictability of Pre-EIA Oil Futures Returns

Panel A	Post-API Return	
	OLS	Robust Regression
Slope	0.17***	0.14***
	(0.04)	(0.05)
Intercept	0.002	-0.01
	(0.02)	(0.02)
R^2	0.04	0.02
Panel B	Predictor	
	OLS	Robust Regression
Slope	-0.06***	-0.05**
	(0.02)	(0.02)
Intercept	-0.0002	-0.01
	(0.02)	(0.02)
R^2	0.02	0.01
Panel C	Post-API Return and Predictor	
	OLS	Robust Regression
Post-API Return	0.16***	0.13**
	(0.06)	(0.05)
Predictor	-0.02	-0.01
	(0.03)	(0.03)
Intercept	0.0001	-0.01
	(0.02)	(0.02)
R^2	0.04	0.02

Panel A reports estimates for the following model: $R_t^{EIA} = \alpha + \beta R_{t-1}^{API} + \varepsilon_t$, where R_t^{EIA} is the oil futures return from 60 min before to 1 min before the EIA announcement and R_{t-1}^{API} is the oil futures return in the 30 minutes after the API announcement. Panel B reports estimates for the following model: $R_t^{EIA} = \alpha + \beta P_t + \varepsilon_t$, where P_t is computed as the difference between the API Actual and the Bloomberg median consensus divided by the level of inventory. Panel C reports the estimates for the following model: $R_t^{EIA} = \alpha + \beta_1 R_{t-1}^{API} + \beta_2 P_t + \varepsilon_t$. The sample period is from January 2011 to August 2019 and contains 385 announcements. The regressions are estimated using (1) OLS with the White (1980) heteroskedasticity consistent covariance matrix and (2) Yohai (1987) MM weighted least squares procedure robust to outliers. Standard errors are shown in parentheses. *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

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Table 3: Predictability of Oil Inventory Surprises

Panel A	Without Lags of Surprise	L=1
Intercept	0.03	0.02
	(0.04)	(0.04)
Post-API Return	-0.92***	-0.95***
	(0.09)	(0.09)
R^2	0.23	0.24
Panel B	Without Lags of Surprise	L=2
Intercept	0.03	0.03
	(0.04)	(0.04)
Predictor	0.68***	0.69***
	(0.03)	(0.04)
R^2	0.51	0.51

Panel A reports the estimates for the following model: $S_t = \alpha + \beta R_{t-1}^{API} + \sum_{n=1}^{L} \gamma_n S_{t-n} + \varepsilon_t$, where S_t is the EIA oil inventory surprise for day *t* and R_{t-1}^{API} is the return of the oil future contracts in the 30 minutes after the API announcement. Panel B reports the estimates for the following model: $S_t = \alpha + \beta P_t + \sum_{n=1}^{L} \gamma_n S_{t-n} + \varepsilon_t$, where P_t is computed as the difference between the API Actual and the Bloomberg median consensus divided by the level of inventory. S_t is computed as the difference between the EIA oil inventory change and Bloomberg median consensus divided by the level of inventory. The optimal number of lags was determined using the Akaike information criterion. The sample period is from January 2011 to August 2019 and contains 385 announcements. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

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Table 4: Predictability of Pre-EIA Oil Futures Return under Different Levels of Liquidity

	Low Liquidity	High Liquidity
Post-API Return	0.20***	0.10
	(0.07)	(0.07)
Intercept	0.01	-0.01
	(0.04)	(0.03)
R^2	0.05	0.01

This table reports estimates for the following model: $R_t^{EIA} = \alpha + \beta R_{t-1}^{API} + \varepsilon_t$ where R_t^{EIA} is the oil futures return from 60 minutes before to one minute before the EIA announcement and R_{t-1}^{API} is the oil futures return in the 30 minutes after the API announcement. We split the sample into times of high and low market liquidity. High is defined as above the median and low is defined as below the median. The sample period is from January 2011 to August 2019 and contains 385 announcements. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

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Table 5: Predicting Returns of Energy Stocks

	CRSP Oil Companies	NYSE ARCA Oil and Gas Index	
Time Interval	Open – Close	9:30 am – 9:40 am	9:40 am – Close
Slope	0.01*** (0.0002)	0.13*** (0.04)	0.13 (0.11)
Intercept	-0.09*** (0.01)	-0.01 (0.02)	-0.03 (0.06)
R^2	0.01	0.03	0.003

The first column reports estimates for the following model: $R_t^{CRSP} = \alpha + \beta R_{t-1}^{API} + \varepsilon_t$, where R_t^{CRSP} is the full day (open-to-close) return of all oil companies in the CRSP database on the days of the EIA announcements and R_{t-1}^{API} is the oil futures return in the 30 minutes after the API announcement. The second and third columns report estimates for the following model: $R_t^{NYSE} = \alpha + \beta R_{t-1}^{API} + \varepsilon_t$, where R_t^{NYSE} is the return of the NYSE ARCA Oil and Gas Index on the days of the EIA announcements in the specified time window. The sample period for the estimations is from January 2011 to August 2019 and contains 385 announcements. The regressions are estimated using OLS with the White (1980) heteroskedasticity consistent covariance matrix. Standard errors are shown in parentheses. *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

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Table 6: Predicting the EIA Announcements using the Bloomberg Consensus and API Actual

	OLS	Robust Regression
Intercept (α)	0.11	-0.04
	(0.13)	(0.14)
BC_t (β_1)	0.38***	0.43***
	(0.07)	(0.07)
AP_t (β_2)	0.69***	0.69***
	(0.03)	(0.04)
R^2	0.65	0.50

This table reports estimates for the following model $EIA\ Actual_t = \alpha + \beta_1 BC_t + \beta_2 AP_t + \varepsilon_t$, where $EIA\ Actual_t$ is the weekly EIA oil inventory change released on day *t*, BC_t is the Bloomberg median consensus forecast for that release, and AP_t is the API actual released in the previous day. The regressions are estimated using (1) OLS with the White (1980) heteroskedasticity consistent covariance matrix and (2) Yohai (1987) MM weighted least squares procedure robust to outliers. The sample period is from January 2011 to August 2019 and contains 385 announcements. Standard errors are shown in parentheses. *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

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Table 7: Response of Oil Futures Prices to Oil Inventory Announcements

	(1) Bloomberg consensus	(2) API actual	(3) Combination Forecast
Panel A		OLS	
Slope	-0.24***	-0.42***	-0.49***
	(0.04)	(0.05)	(0.05)
R^2	0.09	0.18	0.21
Panel B		Robust Regression	
Slope	-0.16***	-0.25***	-0.31***
	(0.03)	(0.03)	(0.03)
R^2	0.06	0.08	0.11
Panel C		Identification-Through-Censoring (ITC)	
Slope	-2.03***	-1.89***	-1.89***
	(0.43)	(0.24)	(0.26)
Pseudo-R^2	0.97	0.95	0.94
Proportion of variance of measured surprise due to noise	88%	78%	74%

This table reports estimates for the following model: $R_t = \alpha + \beta S_t + \varepsilon_t$, where R_t is the oil futures return from 5 minutes before to 10 minutes after the EIA announcement and S_t is the oil inventory surprise. The inventory surprise is computed as the difference between the EIA actual and a market expectation proxy, divided by the inventory level. The inventory surprises are expressed in percentage terms. The Bloomberg consensus forecast, the API actual, and the combination forecast are used to proxy for market expectations in columns (1), (2), and (3), respectively. The combination forecast is computed using the relative importance method proposed by Lindeman, Merenda, and Gold (1980). The regressions are estimated using (1) OLS with the White (1980) heteroskedasticity consistent covariance matrix and (2) Yohai (1987) MM weighted least squares procedure robust to outliers and (3) the identification-through-censoring (ITC) technique proposed by Rigobon and Sack (2008). The sample period is from January 2011 to August 2019 and contains 385 announcements. Standard errors are shown in parentheses. *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

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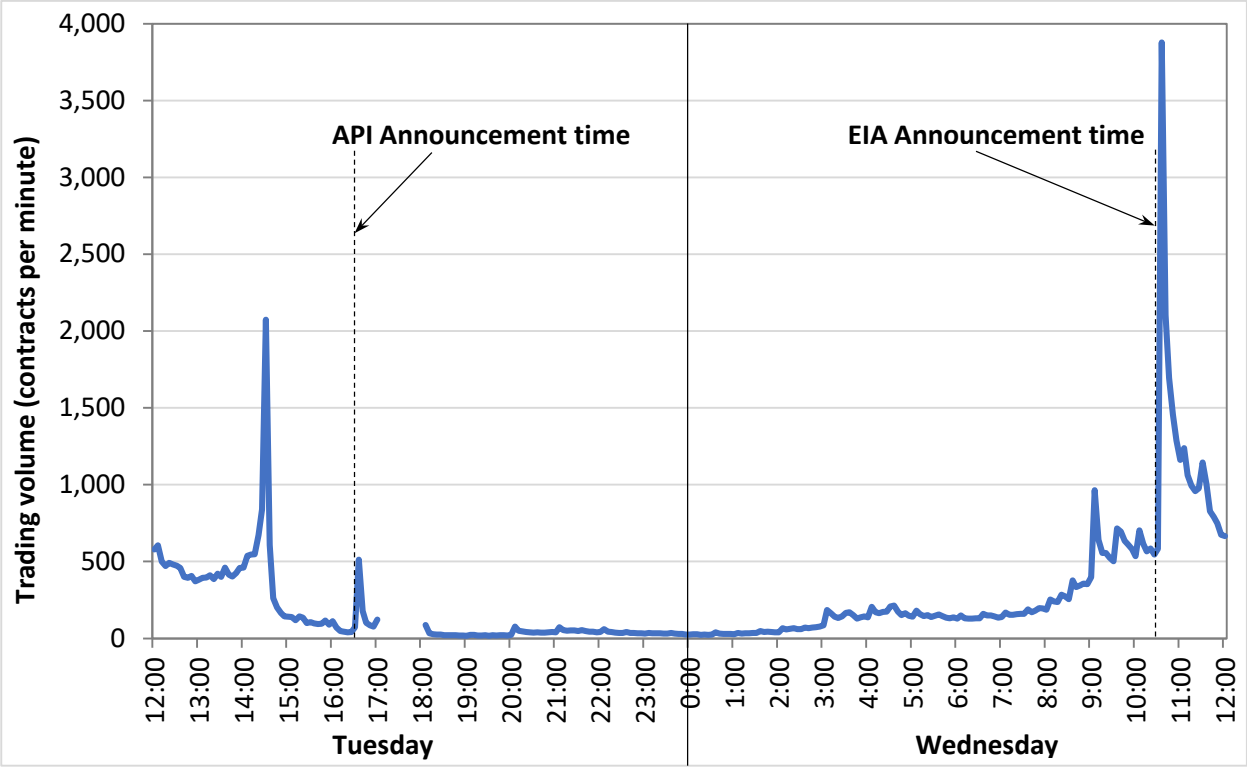
Table 8: Performance of Trading Strategy based on the Predictor

Window	Full Sample (N=385)	Large Values of Predictor (N=77)
(-60 min, -1 min)	0.67	2.10
(-90 min, -1 min)	0.62	1.05
(-120 min, -1 min)	0.54	1.19

This table reports the annualized Sharpe ratio of the following trading strategy: selling (buying) oil futures contracts 60 or 90 or 120 minutes before the EIA announcements and closing the position one minute before the EIA announcement if the predictor is positive (negative). The predictor is computed as the API Actual minus the Bloomberg median consensus. The large values of the predictor are defined as those in the top and bottom deciles. The Sharpe ratio is computed by dividing the sample mean return by the sample standard deviation. The annualized Sharpe ratio is computed based on per-event Sharpe ratio times the square root of the average number of events per year. The sample period is from January 2011 to August 2019 and contains 385 announcements.

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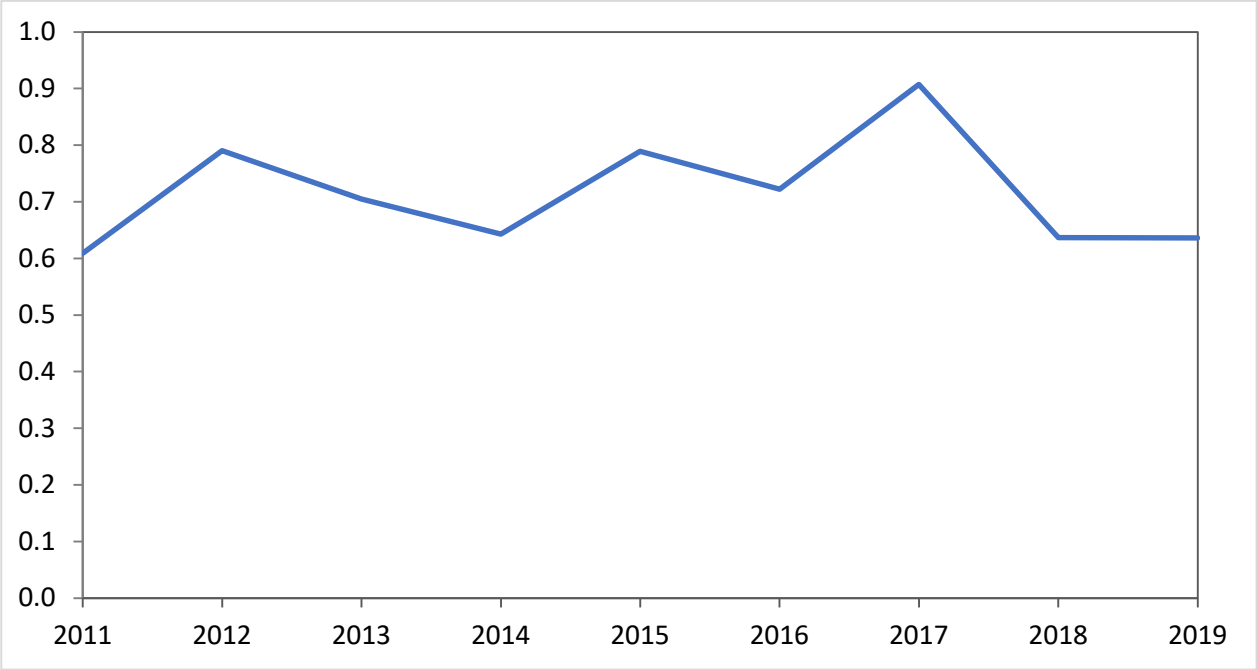
Figure 1: Average Trading Volume around API and EIA Oil Inventory Announcements



The figure shows the average trading volume in the nearby WTI crude futures contracts measured as the number of contracts per minute. The time is stated in U.S. Eastern Time. The sample period is from January 2011 to August 2019 and contains 385 EIA inventory announcements.

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Figure 2: Weights Allocated to the API Actual in the Combination Forecast

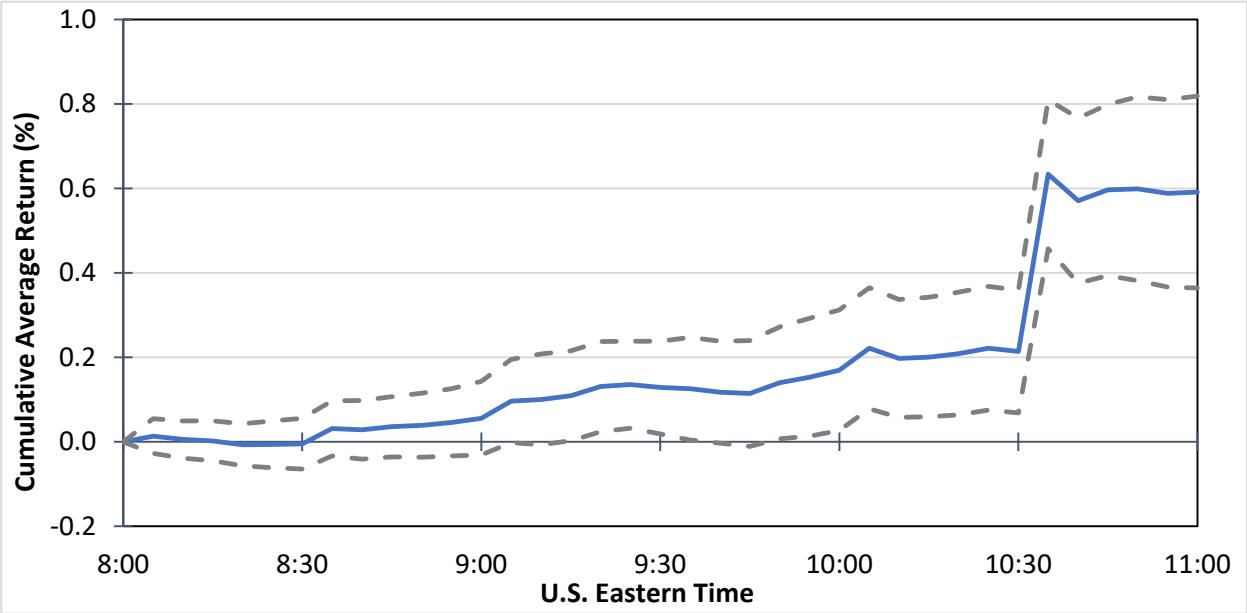


This figure shows the weights allocated to API actuals in constructing the combination forecast based on the relative importance method proposed by Lindeman, Merenda, and Gold (1980) and estimated separately for each year. The remaining weight in the combination forecast is allocated to the Bloomberg consensus forecast. The period is from January 2011 to August 2019 and contains 385 announcements.

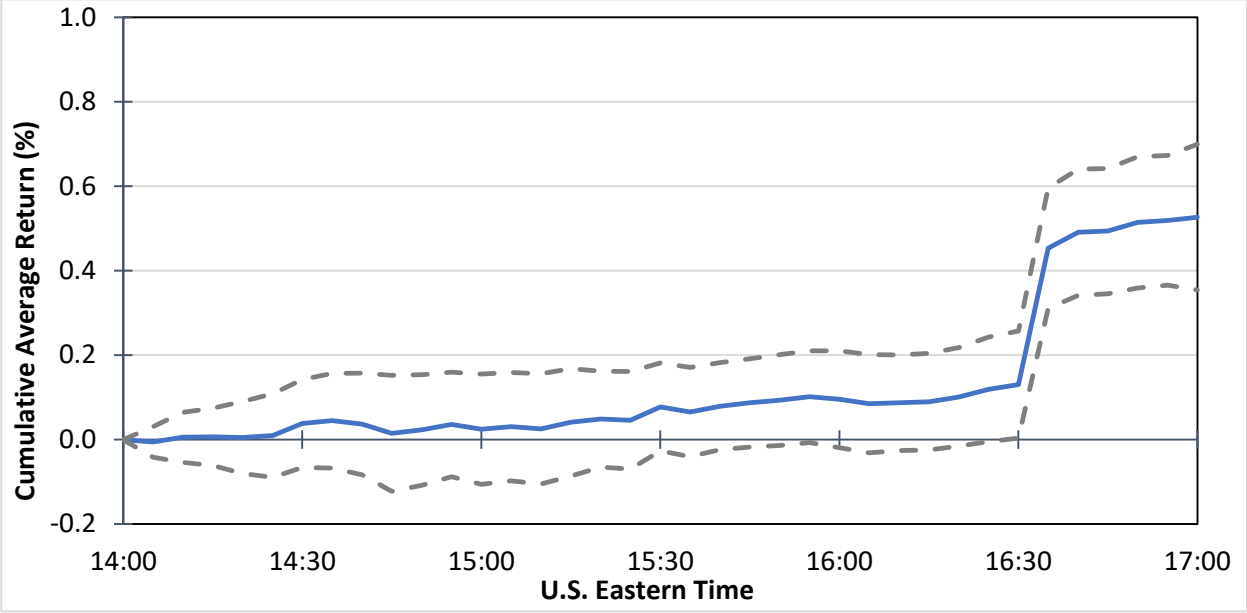
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Figure 3: Cumulative Average Returns around Oil Inventory Announcements

Panel A. EIA Oil Inventory Announcements



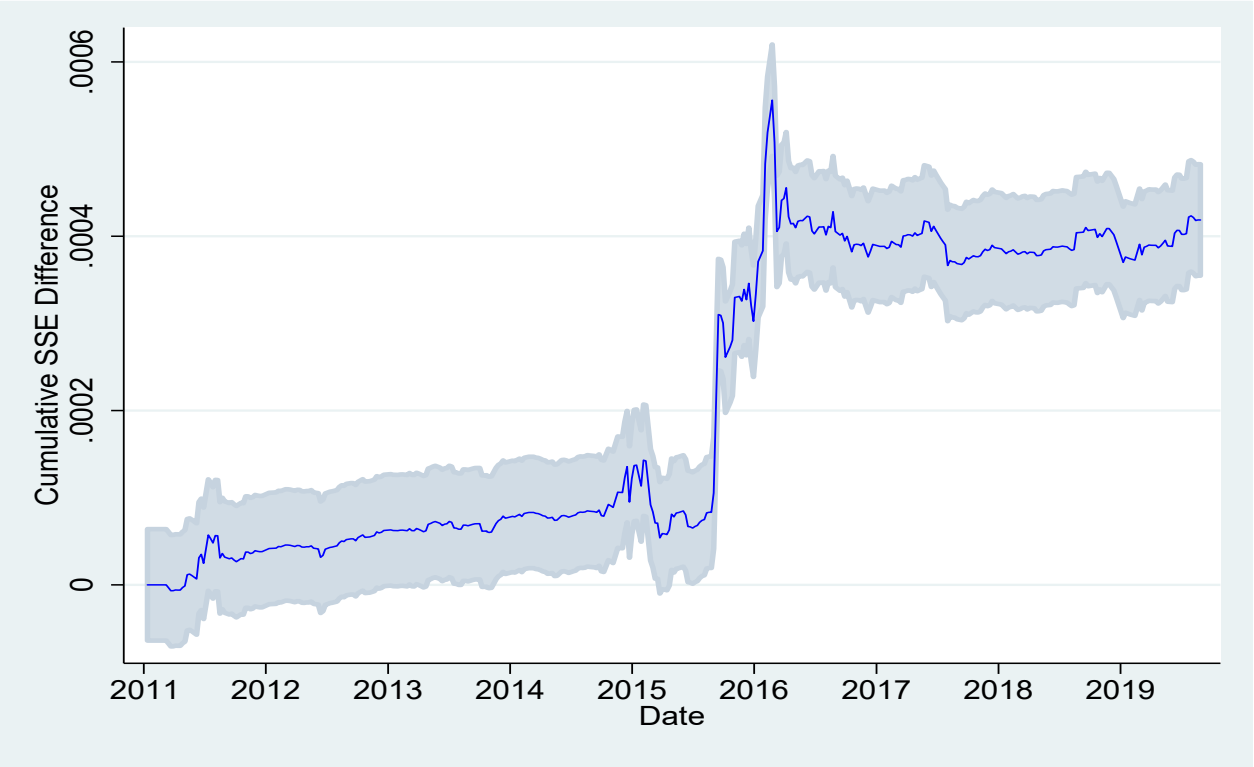
Panel B. API Oil Inventory Announcements



Panel A shows the cumulative average return (CAR) of the nearby oil futures contract around the releases of the weekly Oil Storage Report. Panel B shows the CAR around the API weekly oil inventory announcements. To aggregate CARs for positive and negative surprises, we invert the sign of returns for positive surprises as in Kurov, Sancetta, Strasser, and Wolfe (2019). In Panel A, the inventory surprise is computed as the difference between the EIA actual and the Bloomberg median forecast, divided by the inventory level. In Panel B, the inventory surprise is computed as the difference between the API actual and the Thomson Reuters survey, divided by the inventory level. Only observations with inventory surprises in the top and bottom deciles are used. The sample period is from January 2011 to August 2019 and contains 77 announcements. The time is stated in U.S. Eastern Time. The dashed lines represent 95% confidence intervals.

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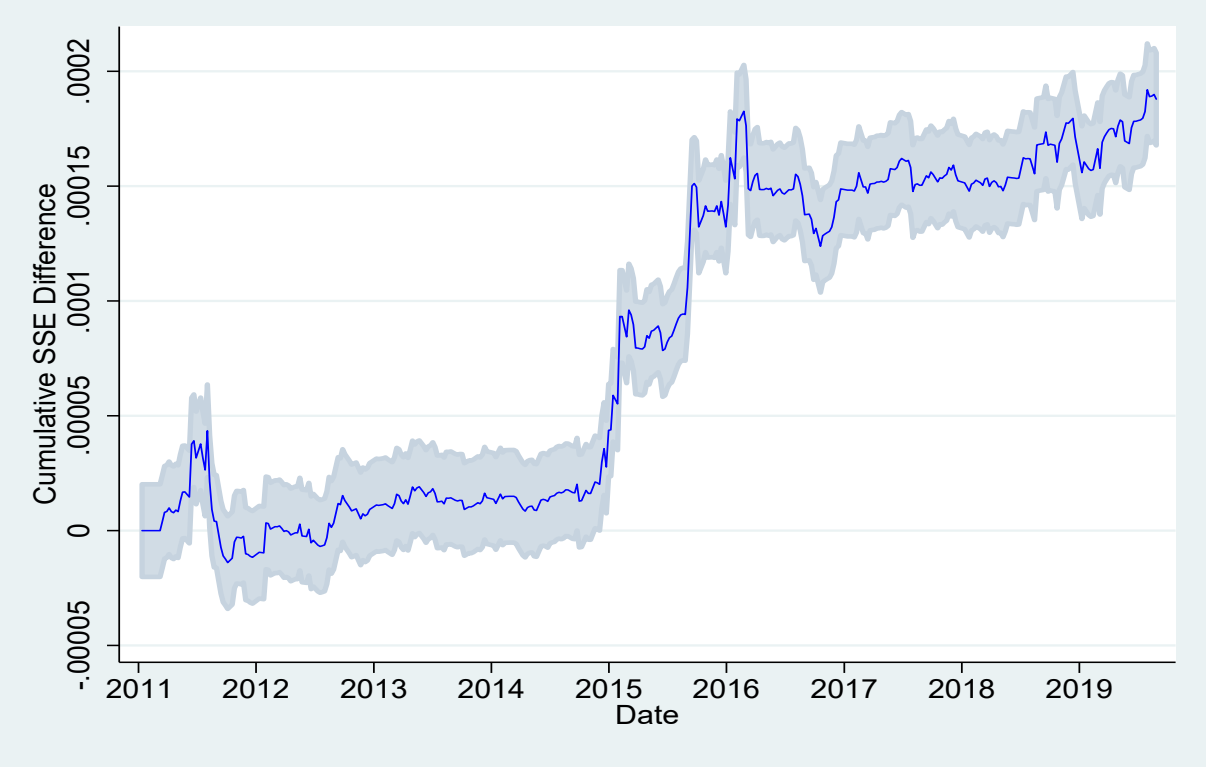
Figure 4: Out-of-Sample Predictability of Pre-EIA Oil Futures Returns using Post-API Oil Futures Returns



This figure plots out-of-sample (OOS) performance of a weekly predictive regression using the methodology proposed by (Welch and Goyal, 2008). These are the cumulative squared predictions errors of the Null minus the cumulative squared prediction errors of the alternative. The alternative model is $R_t^{EIA} = \alpha + \beta R_{t-1}^{API} + \varepsilon_t$, where R_t^{EIA} is the oil futures return from 60 minutes before to one minute before the EIA announcement, R_{t-1}^{API} is the oil futures return in the 30 minutes after the API announcement. The sample period is from January 2010 to August 2019 and contains 385 announcements. We use the data for 2010 to obtain initial parameter estimates for our forecasting model. The Null uses the recursively estimated mean of the pre-EIA oil futures return. An increasing line indicates better performance of the alternative model. A decreasing line indicates better performance of the Null. The blue band represents the equivalent of a 95% confidence interval and is constructed according to the MSE-T critical values from McCracken (2007).

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Figure 5: Out-of-Sample Predictability of Pre-EIA Oil Futures Returns using the Predictor



This figure plots out-of-sample (OOS) performance of a weekly predictive regression using the methodology proposed by (Welch and Goyal, 2008). These are the cumulative squared predictions errors of the Null minus the cumulative squared prediction errors of the alternative. The alternative model is $R_t^{EIA} = \alpha + \beta P_t + \varepsilon_t$, where R_t^{EIA} is the oil futures return from 60 minutes before to one minute before the EIA announcement, and P_t is computed as the difference between the API Actual and the Bloomberg median consensus divided by the level of inventory. The sample period is from January 2010 to August 2019 and contains 385 announcements. We use the data for 2010 to obtain initial parameter estimates for our forecasting model. The Null uses the recursively estimated mean of the pre-EIA oil futures return. An increasing line indicates better performance of the alternative model. A decreasing line indicates better performance of the Null. The blue band represents the equivalent of a 95% confidence interval and is constructed according to the MSE-T critical values from McCracken (2007).

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Appendix

I. *Identification-Through-Censoring*

In a part of our analysis, we use the identification-through-censoring (ITC) technique proposed by Rigobon and Sack (2008). The ITC technique is employed to adjust for the bias that stems from measurement error in the inventory surprises. The measurement error could be due to two reasons. First, our measures of market expectations are imperfect, as they do not incorporate all relevant information available at the time of the announcement. Second, the actual released value of the inventory change is not precise, since it does not cover the entire population of oil companies.[20] We use several proxies for market expectations, which helps to shed some light on the sources of measurement error in the inventory surprises. The effect of inventory news on crude oil futures can be estimated using the following simple regression model:[21]

$$\begin{aligned} R_t &= \gamma z_t + \varepsilon_t \\ z_t &= \Delta I_t - E_{t-\tau}[\Delta I_t] \end{aligned} \tag{A.1}$$

where R_t is the oil futures return computed in the specific intraday window around the announcement, z_t is the inventory surprise calculated as the difference between the actual change of inventory level, ΔI_t, and the proxy for the market expectations of that inventory change, $E_{t-\tau}[\Delta I_t]$. As mentioned above, z_t is measured with error. Therefore, it can be represented as:

$$z_t = z_t^* + \delta_t, \tag{A.2}$$

where δ_t is the measurement error and z_t^* is the "true" inventory surprise. In a regular OLS estimation, the impact of inventory surprises on returns is represented as follows:

[20] The EIA oil inventory survey usually covers about 90% of the population of oil companies, which leaves about 10% not surveyed. This may cause some noise in the estimates of the change in oil inventory. This issue is discussed at https://www.api.org/products-and-services/statistics/api-weekly-statistical-bulletin.

[21] For a more detailed discussion of the ITC technique in the context of energy inventory announcements, see Halova et al. (2014).

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$$R_t = \gamma z_t + v_t,$$
$$v_t = \varepsilon_t - \gamma \delta_t. \tag{A.3}$$

However, the coefficient γ is biased since the regressor is correlated with error term. We can express the OLS estimate of the response coefficient as:

$$\hat{\gamma}_{OLS} = \gamma \left(1 - \frac{\sigma_\delta^2}{\sigma_{z^*}^2 + \sigma_\delta^2}\right). \tag{A.4}$$

This means that the OLS estimator is biased downwards. Rigobon and Sack (2008) argue that dealing with the measurement error bias presents an identification problem. They suggest this problem can be solved by noting that the measurement error is "censored" in periods with no announcements. In such periods both the "true" inventory surprise z_t^* and the measurement error δ_t are zero. Using returns in the same intraday interval on the day before the announcement, R_{t-1}, the model parameters can be estimated using the generalized method of moments (GMM) based on the following set of moment conditions:

$$var(R_{t-1}) = \sigma_\varepsilon^2,$$
$$var(R_t) = \gamma^2 \sigma_{z^*}^2 + \sigma_\varepsilon^2,$$
$$var(z_t) = \sigma_{z^*}^2 + \sigma_\delta^2, \tag{A.5}$$
$$cov(R_t, z_t) = \gamma^2 \sigma_{z^*}^2.$$

Solving these equations produces the following estimator of the market response coefficient:

$$\gamma = \frac{var(R_t) - var(R_{t-1})}{cov(R_t, z_t)}. \tag{A.6}$$

For returns on non-announcement days, we use the same intraday interval (from 10:25 am to 10:40 am) as the interval used to measure the returns around the announcement.

We compute the following pseudo-R^2 statistic to compare the explanatory power of the

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ITC estimation with that of the OLS:

$$\text{pseudo-}R^2 = 1 - \frac{\sigma_\varepsilon^2}{var(R_t)} \quad \text{(A.7)}$$

Intuitively, the pseudo-R^2 captures the fraction of return variance explained by the model. We estimate the variance of the structural shocks ε_t using non-announcement returns. Therefore, equation (A.7) assumes that inventory surprises fully explain the post-announcement increase in the variance of returns.

II. *Out-of-Sample Forecasting*

Several studies show that the predictors that seem to perform well in-sample may lose their predictive power out-of-sample (e.g., Welch and Goyal 2008). Hence, we conduct an out-of-sample predictability analysis for our main empirical findings. We use the same methodology as in Welch and Goyal (2008). The benchmark model (Null) is the recursively estimated mean of the pre-EIA oil futures return. Subsequently, we compare the benchmark model with the following alternative predictive regression:

$$Y_t = \alpha_0 + \alpha_1 X_{t-1} + \varepsilon_t \quad \text{(A.8)}$$

where X_{t-1} is either the post-API oil futures return or the predictor, which is computed as the difference between the API actual and the Bloomberg consensus forecast divided by the level of inventory. The performance of the predictive regression is evaluated by computing the difference between the cumulative squared predictions errors of the Null and the cumulative squared prediction errors of the Alternative. If this difference is positive, it means the Alternative performs better than the Null, and if it is negative it means the Null has better predictive ability than the Alternative.

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